Related Work and Gap Analysis: sEMG Gesture Recognition with GNNs

No	Paper	Data and graph	Model and evaluation	Key results	Limitations and gap
1	Vipulanandan et al., Real-Time Gesture Recognition with a GNN on sEMG (arXiv 2026)	Data: Myo band, 8 ch, 8 subjects, 5 gestures. Graph: node = electrode, edge = Pearson correlation per window	Simple 2-layer GNN. Random 20% of overlapping windows as test	98.19% to 99.89%. 48 ms per prediction	only 5 gestures, no cross-subject test; random overlapping split may inflate accuracy. It uses a repetition based split on DB2 and add cross-subject results
2	Hu et al., MFN-SPSCA (IEEE TNSRE 2024)	Data: Myo sEMG and accelerometer; MyoData (17 subj, 8 ch), NinaPro DB5 (10 subj, 16 ch). Graph: node = channel, edge = thresholded Pearson correlation and spatial partitioning	Multi-branch CNN/BiLSTM ,ST-GCN,CWT, cross attention. DB5 reps 1,3,4,6 train, 2,5 test	98.78% MyoData, 91.60% DB5. 12 to 21 ms, about 37 MB	Known users only, no cross-subject; heavy model, needs accelerometer. It used dataset is DB5 not DB2; try a lighter sEMG only GNN with leave-subject out evaluation
3	Xu et al., CNN-MSTGCN (IEEE TNSRE 2024)	Data: own HD sEMG, 11 subj, 17 gestures, 128 ch. Graph: node = electrode, edge = grid distance and learned functional connection	CNN and MSTGCN with spatial temporal attention. User-specific 7:3, user independent 11-fold, transfer learning	96.9% user specific. 68.0% user independent (ResNet50 47.5%, LSTM 57.1%). 92.3% with transfer learning	Offline only, 68% not satisfactory, needs new-user data, no public dataset. It evaluates on public 40 subject DB2, report within subject and cross-subject
4	Lee et al., Spatiotemporal GNNs for Generalizable and Interpretable EMG	Data: BandMyo, NinaPro DB5, UCI, CapgMyo A and B. Graph: node = sensor, edge = physical proximity (KNN 1, KNN-2, fully	GCN (also GAT, GraphLSTM, STGCN, GCLSTM). Random 80:20 per subject; intra-	Intra-subject 99.3 to 99.9%. Inter-subject 92.4% DB5, 98.9% UCI, 99.4% BandMyo. 28	Data has been randomly splitted may be optimistic, inter subject protocol unclear, no latency (our note). It do not uses DB2. It uses a clear protocol, compare graph types,

	Classification (IEEE TNSRE 2025)	connected) and temporal links between windows	and inter-subject	sensors give about 97%	analyse channel importance
5	Massa et al., GNNs for HD EMG Movement Intention Recognition (IEEE RASSE 2022)	Data: HD-EMG, 20 subj, 65 gestures, 128 ch. Graph: node = channel (65 raw samples), edge = grid distance (128 nodes, 884 edges)	EMG-GNN: 6 SAGEConv + mean pooling + MLP. Each subject separately, 60/20/20 shuffled graphs	Mean error 8.75% (SD 4.92). 45 of 65 gestures under 10% error	The model is weak on gestures with 2+ degrees of freedom; initial study, one dataset, no cross-subject test, no latency. It has a gap on per-class confusion analysis on DB2 and comparison with published baselines
6	Zhou et al., CovGCN (IEEE Sensors Journal 2025)	Data: NinaPro DB2 (40 subj, 12 ch, Ex. B/C/D) and own UOW (8 ch, 3 limb positions). Graph: node = channel, predefined topology refined per gesture by covariance attention	7 graph-conv blocks, 1.07 M parameters. DB2 reps 1,3,4,6 train, 2,5 test, per subject	DB2 Ex. B/C/D: 88.02 / 81.30 / 93.30%. UOW 84.69%. 29.6 ms	Only on specific subject , able-bodied, offline; Transformer slightly higher (88.91% on Ex. B).